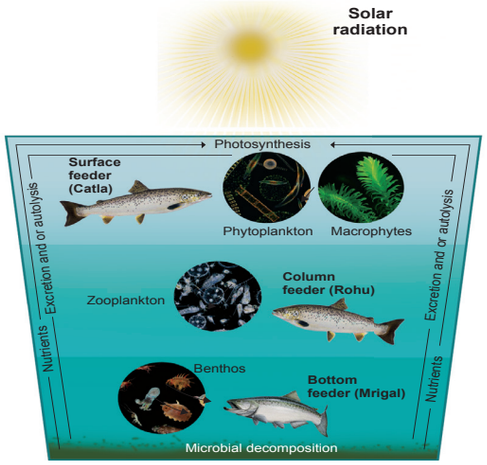


Fig.11.1 Types of feeders (niche) in pond ecosystem

11.3 Major Abiotic Components or Factors

The abiotic factors include the chemical and physical factors which influence or affect organisms and their functioning in their environment. The common abiotic factors are:

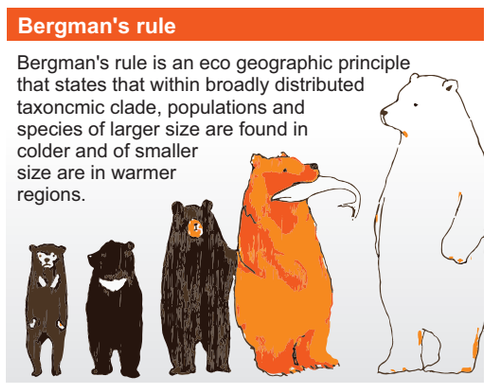
Temperature

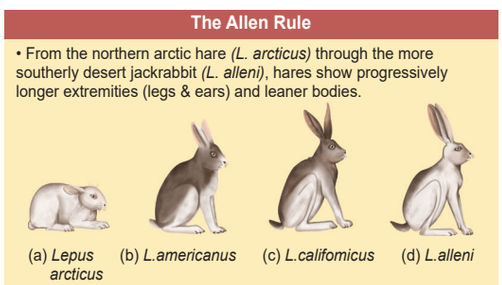
Temperature or degree of hotness and coldness is an essential and variable factor in any environment. It influences all forms of life by affecting many vital activities of organisms like metabolism, behaviour, reproduction, development and even death in the Biosphere. The minimum and maximum temperature of an environment regulates the survival of a cell.

Van't Hoff's rule

Van't Hoff proposed that, with the increase of every 10°C, the rate of metabolic activity doubles or the reaction rate is halved with the decrease of 10°C. This rule is referred as the Van't Hoff's rule. The effect of temperature on the rate of reaction is expressed in terms of temperature coefficient or Q_{10} value. The Q_{10} values are estimated taking the ratio between the rate of reaction at X°C and rate of reaction at (X-10°C). In the living system the Q_{10} value is about 2.0. If the Q_{10} value is 2.0, it means 10°C increase and the rate of metabolism doubles.

The metabolism of organisms is regulated by enzymes which are temperature sensitive. In many organisms, determination of sex and sex ratio, maturation of gonads, gametogenesis and reproduction is influenced by temperature. In certain environments, the size and colouration of animals are influenced by temperature. Birds and mammals attain greater body size in colder regions than warmer regions (**Bergmann's rule**). Warm blooded animals, living in colder climates, tend to have shorter limbs, ears and other appendages when compared to the members of the same species in warmer climates (**Allen's rule**). In some aquatic environments, an inverse relationship between water temperature and fish meristic characters is observed - lower the temperature, more the vertebrae (**Jordon's rule**).





Temperature influences the distribution of organisms. The tropics have higher diversity and density of populations, when compared to temperate and polar regions.

Adaptations to temperature

Adaptation to temperature is essential for the survival of the species/organisms. Organisms which can survive a wide range of temperature are referred to as **Eurytherms**

XII Std Zoology Chapter 11 EM.indd 178 23/12/2022 16:51:09